Solutions Guide

I chose 4 questions to provide solutions for. These include a hill, a vigenere, an affine, and a morbit cipher.

Solution 1: Hill Decode

---

16) **[300 points]** Decode this hill cipher.

$$\begin{pmatrix} H & O \\ W & D \end{pmatrix} \equiv \begin{pmatrix} 7 & 14 \\ 22 & 3 \end{pmatrix}$$

I	L	Y	H	A	N	I	N	U	M

The matrix can be thought of as: (a b)
(c d)

We first need to compute the inverse matrix of (7 14)
(22 3).

We can use the formula $(ad - bc)^{-1} \cdot \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

We get $(7 * 3 - 14 * 22)^{-1} \cdot \begin{pmatrix} 3 & -14 \\ -22 & 7 \end{pmatrix}$

$(7 * 3 - 14 * 22) = -287 \rightarrow -287 \bmod 26 \rightarrow 25$

The modular inverse for 25 is also 25 (as given in the reference tables) and we can substitute this value into the formula.

25 • (3 -14) *mod 26* → (25 • 3 25 • -14) → (75 mod 26 -350 mod 26) → **(23 14)**
(-22 7) (25 • -22 25 • 7) (-550 mod 26 175 mod 26) **(22 19)**

Now that we have the inverse matrix, we can decode normally.

(23 14) • (I) → (23 14) • (8) → (23 • 8 + 14 • 11) → (338) mod 26 (0) → **(A)**
(22 19) • (L) → (22 19) • (11) → (22 • 8 +19 • 11) → (385) mod 26 (21) → **(V)**

(23 14) • (Y) → (23 14) • (24) → (23 • 24+14 • 7) → (650) mod 26 (0) → **(A)**
(22 19) • (H) → (22 19) • (7) → (22 • 24 + 19 • 7) → (661) mod 26 (11) → **(L)**

(23 14) • (A) → (23 14) • (0) → (23 • 0 +14 • 13) → (182) mod 26 (0) → **(A)**
(22 19) • (N) → (22 19) • (13) → (22 • 0 + 19 • 13) → (247) mod 26 (13) → **(N)**

(23 14) • (I) → (23 14) • (8) → (23 • 8 +14 • 13) → (366) mod 26 (2) → **(C)**
(22 19) • (N) → (22 19) • (13) → (22 • 8 + 19 • 13) → (423) mod 26 (7) → **(H)**

(23 14) • (U) → (23 14) • (20) → (23 • 20 +14 • 12) → (628) mod 26 (4) → **(E)**
(22 19) • (M) → (22 19) • (12) → (22 • 20 + 19 • 12) → (668) mod 26 (18) → **(S)**

That gives us our final answer: AVALANCHES

Solution 2: Vigenere Cryptanalysis

18) **[300 points]** Decode this Vigenere that starts with "Learn"

P	M	N	J	G	J	Z	B	Q	G	R	K	M	I	Z	Q	E	G	Y	A	O	I	N	B	V	B	B	V	T	C
P	B	T	M	S	G	K	X	W	Z	S	Z	E	G	P	X	P	R	M	U	C	G	K	X	I	A	X	B	U	A
G	K	Q	F	R	W	G	L	H	W	B	B	T	Y	H	W	L	X	Q	B	R	Q	A	Y						

We first have to find the keyword, as it is not given.

We are given:
P M N J G J Z B Q G
L E A R N F R O M ?

The solution is to solve in a reverse fashion to find the keyword, and so we subtract the ciphertext and plaintext.

P = 15 → 15 - 11 → 4 → **E** M = 12 → 12 - 4 → 8 → **I** N = 13 → 13 - 0 → 13 → **N**
L = 11 E = 4 A = 0

J = 9 → 9 - 17 $_{(\text{mod } 26)}$ → 18 → **S** G = 6 → 6 - 13 → 19 → **T** J = 9 → 9 - 5 → 4 → **E**
R = 17 N = 13 F = 5

Z = 25 → 25 - 17 → 8 → **I** B = 1 → 1 - 14 → 13 → **N** Q = 16 → 16 - 12 → 4 → **E**
R = 17 O = 14 M = 12

The keyword seems to repeat, so we can assume that it is **EINSTEIN.**

We can then decrypt the rest of the cipher normally by subtracting the keyword from the ciphertext.

EX:
E I N S T E I N E I
P M N J G J Z B Q G
L E A R N F R O M ?

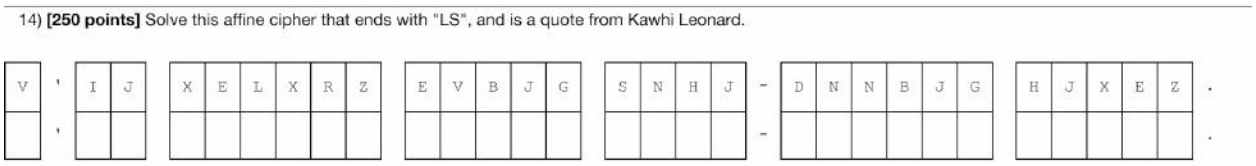
G = 6, I = 8 → 6 - 8 → -2 (mod 26) → **24 (Y)**

We can then use this method to decrypt the rest of the cipher.

Learn from yesterday, live for today, hope for tomorrow. The important thing is not to stop questioning.

Solution 3: Affine Cryptanalysis

We can first set up two equations from the given information.



11a + b = 4 (mod 26) cause L → E
18a + b = 25 (mod 26) cause S → W

Subtract the first equation from the second and we get:
7a = 21 (mod 26)
a = 3 (mod 26)

a can only be 3, so now we can plug it back in the linear equations to get:
11(3) + b = 4 (mod 26) → 33 + b = 4 + (26 * 2) → 33 + b = 56 → **b = 23**

We can then solve the rest of the cipher normally using the formula:

$D(x) = a^{-1}(x - b) \bmod 26$

Ex: (Solving for first three letters)
Inverse of 3 is 9
V → 9 • (21 - 23) → 9 • (-2) mod 26 → -18 mod 26 → 8 → **I**
I → 9 • (8 - 23) → 9 • -15 mod 26 → -135 mod 26 → 21 → **V**
J → 9 • (9 - 23) → 9 • -14 mod 26 → -126 mod 26 → 4 → **E**

We have deduced that the first term is “I’ve”. Using this method, we can solve for the rest of the cipher, and we get:

I’ve always liked home-cooked meals.

Solution 4: Morbit Cryptanalysis

19) **[300 points]** Decipher this quote encoded using the Morbit cipher. You are told it ends with "CRAZY"

```
7 9 5 6 2 6 7 6 5 7 8 7 9 7 8 6 5 8 4 1 3 4 1 9 6

7 7 1 6 6 1 3 7 4 9 4 2 8 2 1 5 9 4 4 8 2 1 5 1 4 3

4 3 1 5 8 3 4 3 3 5 3 7 3 6 5 8 3 4 3 8 6 2 6 5 3

7 3 3 8 5 3 2 6 4 3 3 1 6 8 1 6 5 4 9 4 3 5 8 3 4 4

1 1 4 9 5 6 2 6 7 6 5 7 8 7 9 4 3 5 3 7 1 7 6 3 8 5 6

2 6 7 6 8 5 3 5 3 5 8 4 3 4 2
```

After filling out the morse code and clue given, we can tell that 2 → --, 4 → -•, 3 → •x, 8 → x-, 5 → •-. The clue given tells us that 6 = -x. Filling out the last line, we get

```
2  6  7  6  8  5  3  5  3  5  8  4  3  4  2
--  -x    -x  x-  •-  •x  •-  •x  •-  x-  -•   •x  -•  --
?    ?   /  C       R    A    Z      Y
```

7 can only be xx, x•, or ••.

We can use the process of elimination and guess and check in this case.

If 7 were to equal ‘xx’, that would leave us with 3 x’s in a row, which isn’t possible.

```
2  6  7  6  8  5  3  5  3  5  8  4  3  4  2
--  -x  xx   -x  x-  •-  •x  •-  •x  •-  x-  -•   •x  -•  --
?    ?   /  C       R    A    Z      Y
```

If 7 were to equal ‘x•’, that would leave us with this:

```
5  6  2  6  7  6  8  5  3  5  3  5  8  4  3  4  2
•-  -x  --  -x x•  -x x-  •- •x  •- •x •- x- -• •x -• --
Y   O     /  A  /  C     R    A    Z     Y
```

This would make sense in the middle of a sentence, but considering the sentence ends with this, it is unreasonable to deduce that the phrase would end with ‘a crazy’

If 7 were to equal ‘••’, that would leave us with this:

```
5  6  2  6  7  6  8  5  3  5  3  5  8  4  3  4  2
•-  -x  --  -x ••  -x x-  •- •x  •- •x •- x- -• •x -• --
?   O     U   /  C     R    A    Z     Y
```

This looks like it makes sense, and hence we can conclude that 7 is ••.

Now, we only have ‘xx’ and ‘x•’ to map to the two remaining numbers, 1 and 9.

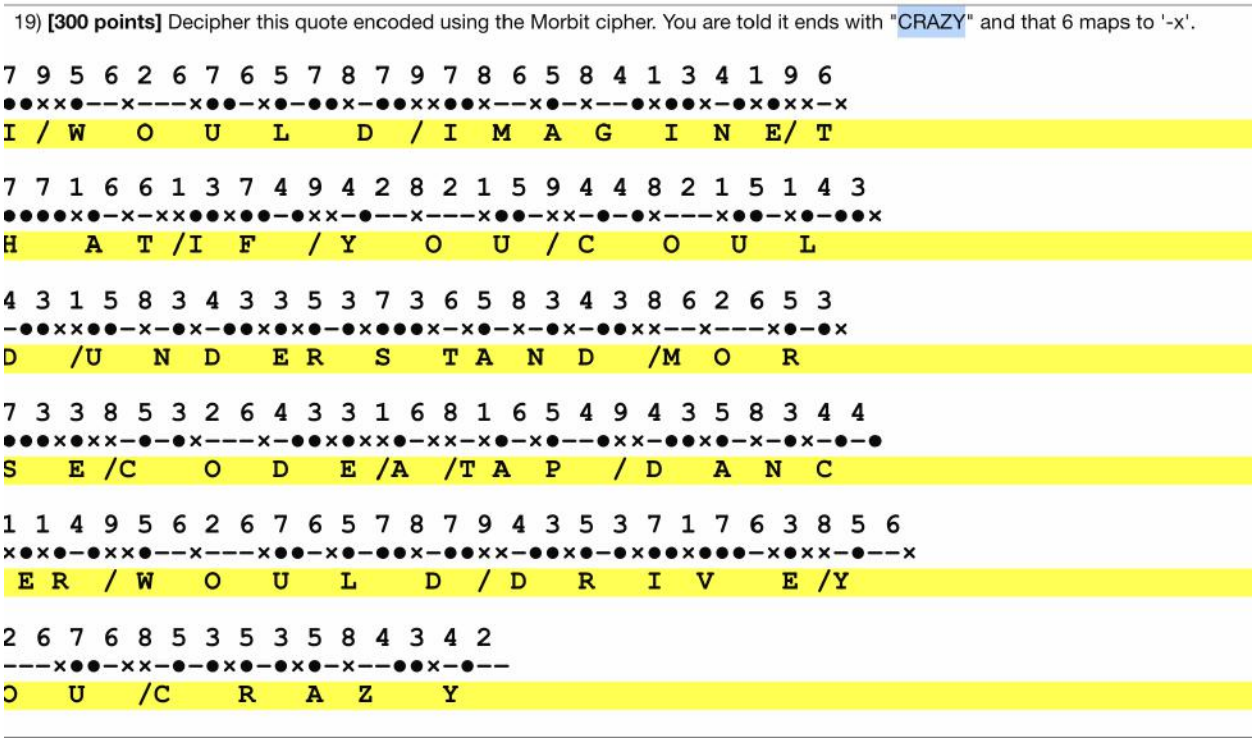
```
7  1  7  6  3  8  5  6  2  6  7  6  8  5  3  5  3  5  8  4  3  4  2
••     •• -x •x x-  •-  -x  --  -x ••  -x x-  •- •x  •- •x •- x- -• •x -• --
          E  /   Y   O    U   /   C     R    A    Z     Y
```

If 1 = ‘xx’, we get:

```
7  1  7  6  3  8  5  6  2  6  7  6  8  5  3  5  3  5  8  4  3  4  2
••  xx •• -x •x x-  •-  -x  --  -x ••  -x x-  •- •x  •- •x •- x- -• •x -• --
    / U   E  /   Y   O    U   /   C     R    A    Z     Y
```

‘UE’ isn’t a word, so that leaves us to assume that 1 is ‘x•’. 9 must then be ‘xx’. We then can fill in the rest of the cipher with the corresponding symbols.

We then get the final answer:



I would imagine that if you could understand morse code a tap dancer would drive you crazy.